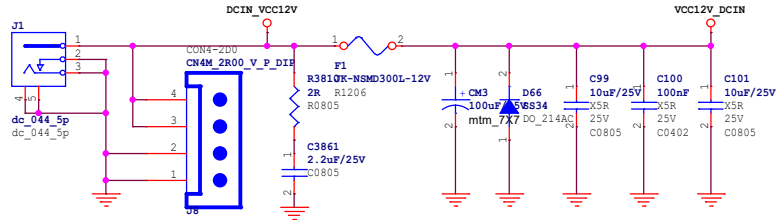
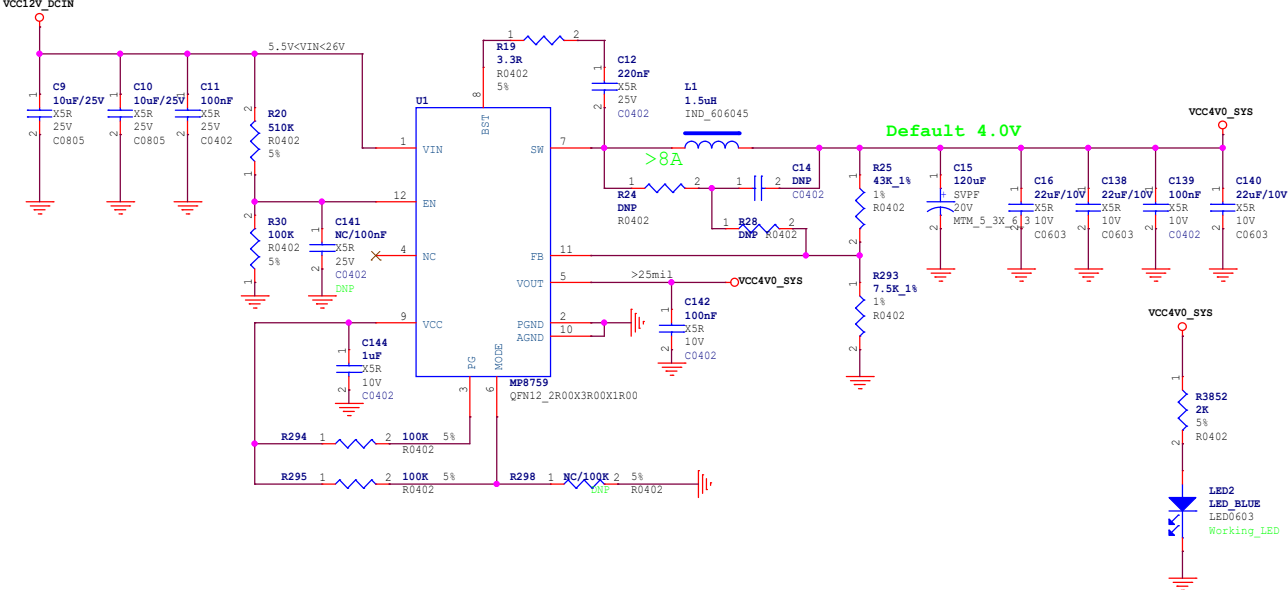


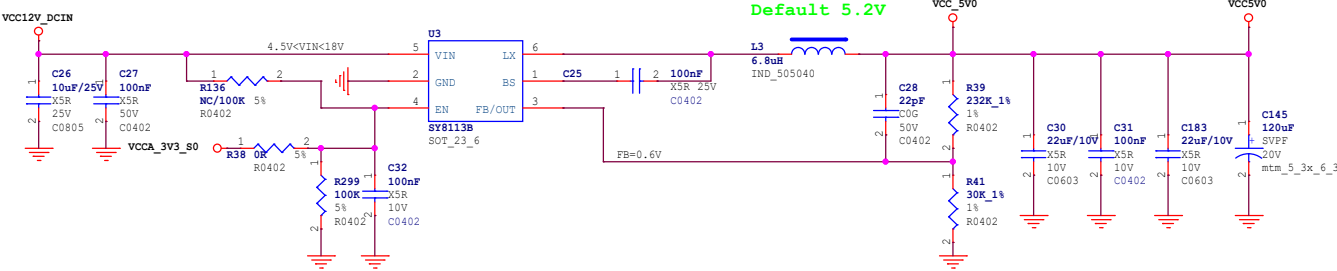
12V/2A DCIN



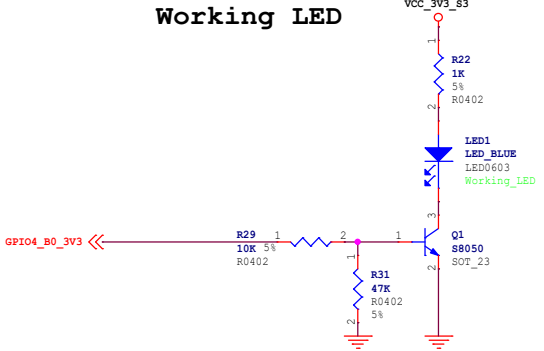
VCC5V0_SYS



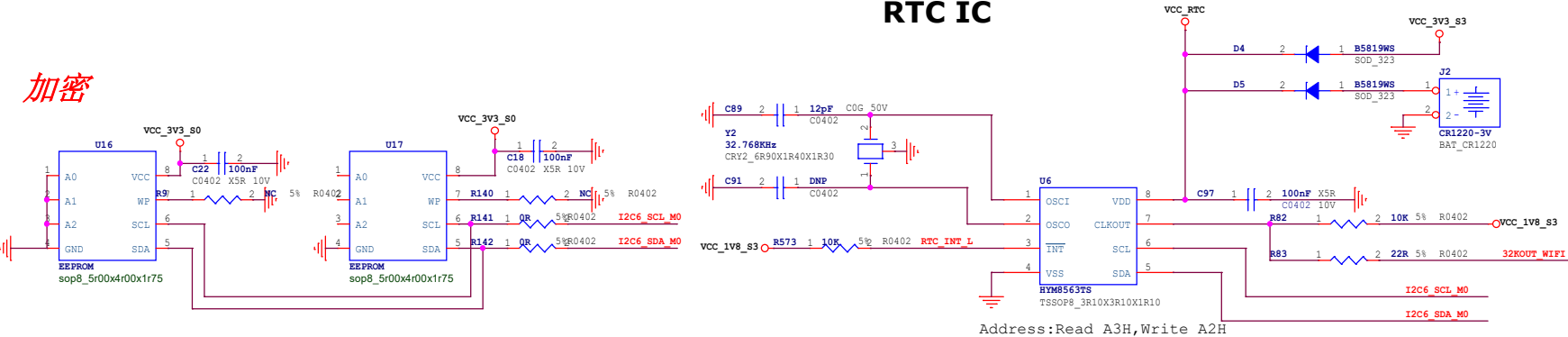
VCC5V0



Working LED



RTC IC



 瑞芯微电子		Rockchip Electronics Co., Ltd	
Project:		RK3588_MXM314_60X82_DB	
File:		02.Power_DC IN	
Date:	Tuesday, September 09, 2025	Rev:	V1.0
Designed by:	Zhangtz	Reviewed by:	Default
		Sheet:	20 of 50

CAMERA控制口

M.2 KEY_B_M

M.2 KEY_B_M

PCIe WIFI6

SATA

以太网GMII

双目2lane*2

MIPI显示0

MIPI显示1

蓝牙UART

TYPEC/DP1

HDMI OUT / EDP

HDMI OUT / EDP

TF卡

GPIO14_0_NIRQ0

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GPIO14_0_NIRQ0

PCIe WIFI6控制口
camera控制口

Codec I2S0控制口

PCIE以太网

WIFI控制口
UART CAN I2C

以太网GMII

双目2lane*2

摄像头4lane

摄像头4lane

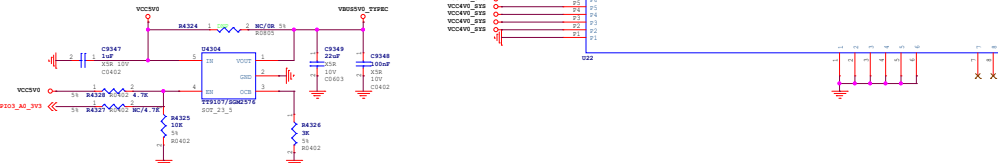
USB HOST

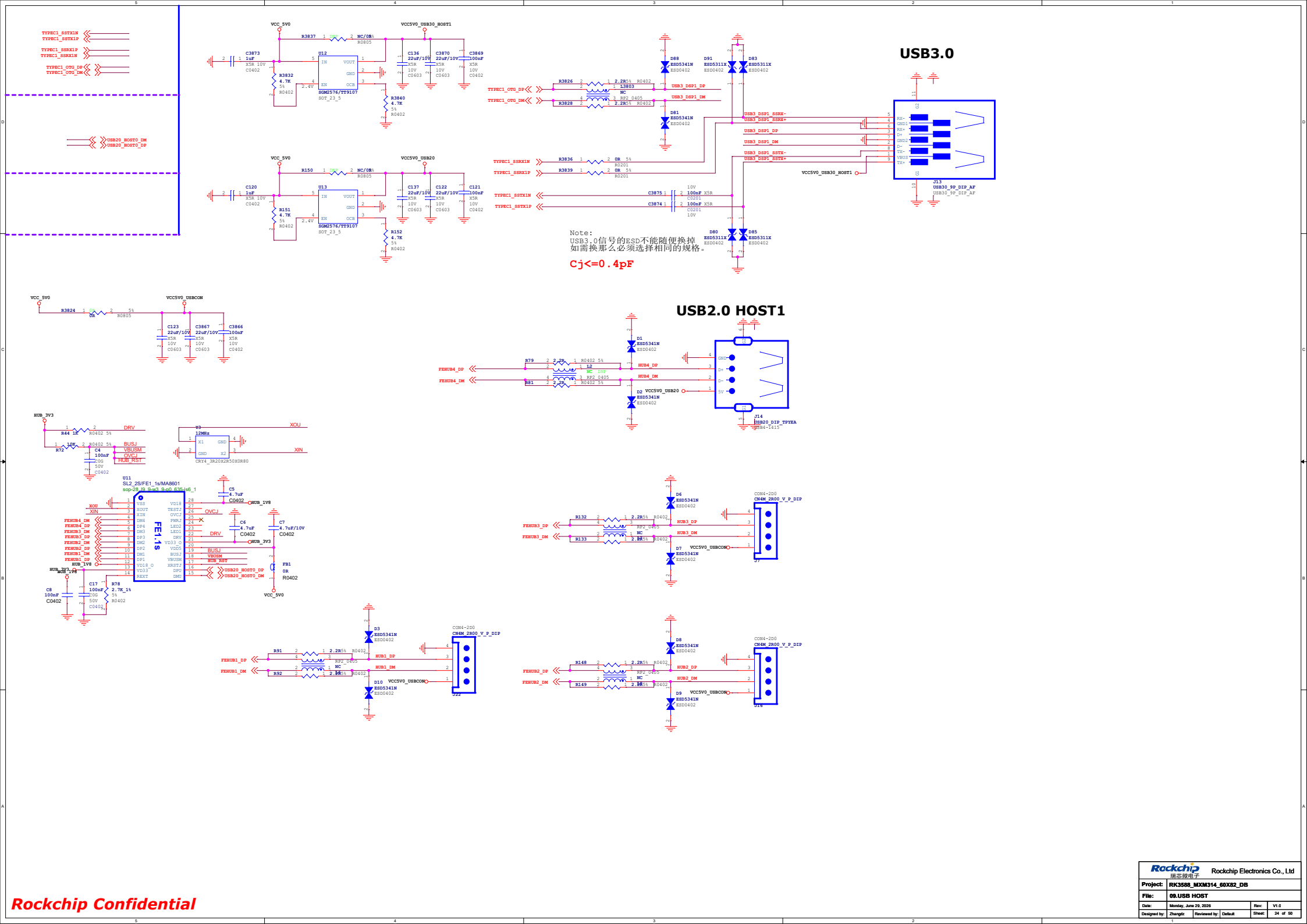
HDMI IN

UART CAM 等等控制口

PCIe控制口

HDMI OUT和 IN 控制口



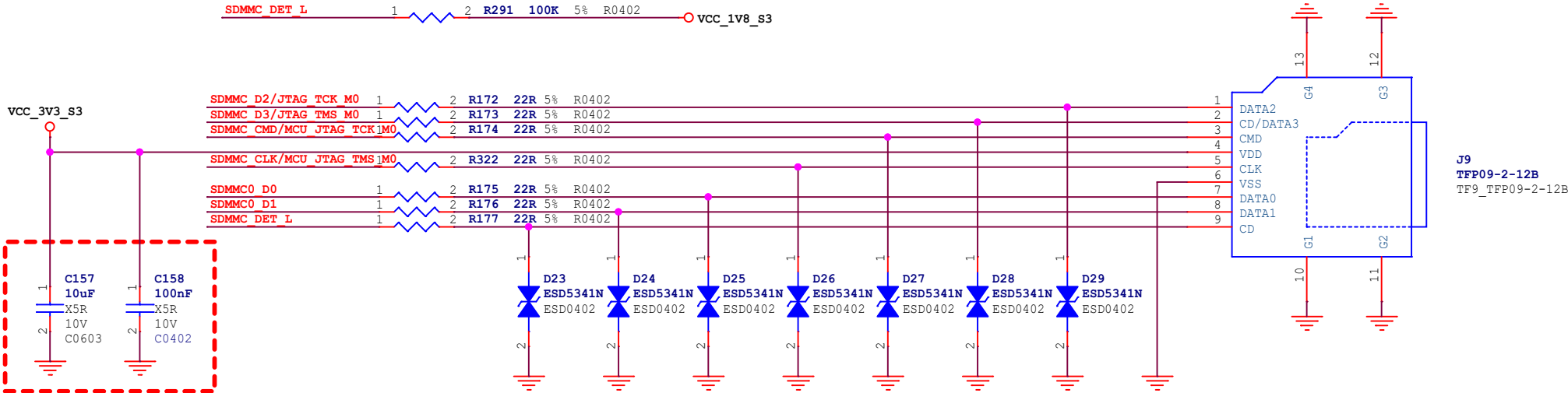


USB3.0

USB2.0 HOST1

Note:
USB3.0 信号的ESD不能随便换掉
如需换那么必须选择相同的规格。
Cj<=0.4pF

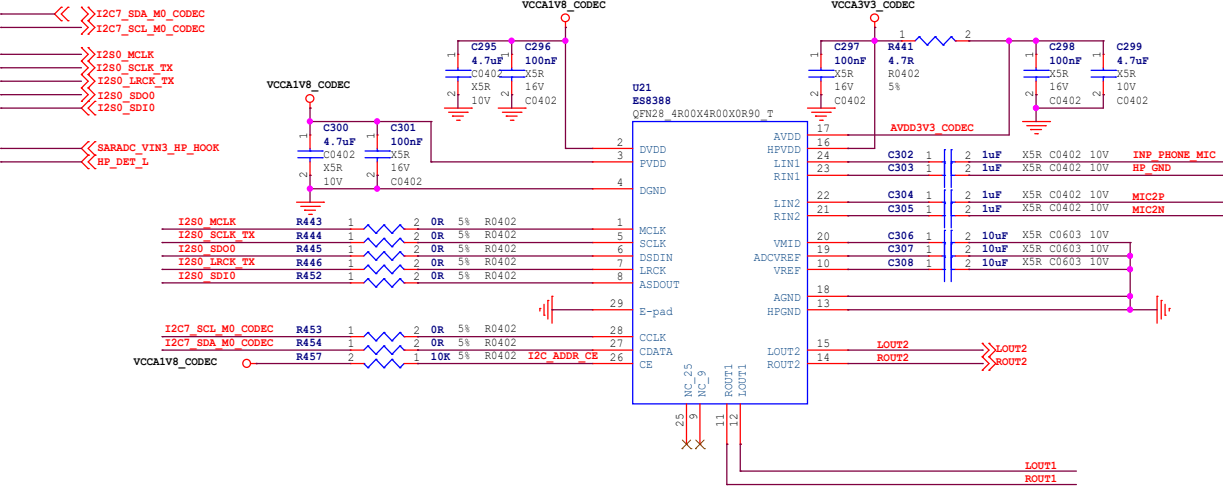
SDMMC D2/JTAG TCK M0
SDMMC D3/JTAG TMS M0
SDMMC CMD/MCU JTAG TCK M0
SDMMC CLK/MCU JTAG TMS M0
SDMMC_DET_L



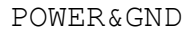
Close to MicroSD Card

MicroSD Card

CODEC ES8388



VCC12V_PA VCC12V_PA VCC12V_PA **AUDIO_AnalogAmp**



SATA30_0_TXP
SATA30_0_TXN
SATA30_0_RXN
SATA30_0_RXP

SATA3.0 Port0

SATA30_0_TXP
SATA30_0_TXN

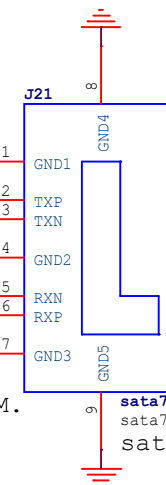
C336 1 2 10nF X7R 10V C0201
C337 1 2 10nF X7R 10V C0201

SATA30_TXP
SATA30_TXN

SATA30_0_RXN
SATA30_0_RXP

C338 1 2 10nF X7R 10V C0201
C339 1 2 10nF X7R 10V C0201

SATA30_RXN
SATA30_RXP



NOTES: The SATA differential trace impedance is 100 OHM.
The SATA trace length is less than 5 inch.

sata7_DIP
sata7_DIP
sat3m3232xxxxx

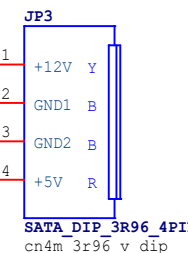
VCC12V_DCIN

VCC12V_SATA

FB8
330R-100M
2.5A
L1206

C405
22uF/25V
X5R
25V
C0805

C116
100nF
X5R
25V
C0402



NOTES: Close to DC Port

VCC5V0

VCC5V0_SATA

FB12
330R-100M
2.5A
L1206

C406
22uF/10V
X5R
10V
C0805

C347
100nF
X5R
10V
C0402

[illegible][illegible]

4G/5G SIM CARD

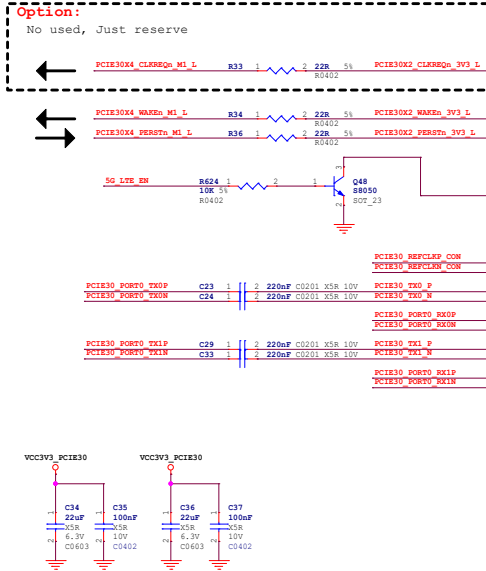
The schematic diagram illustrates the electrical connections for a 4G/5G SIM card module. Key components and connections include:

- Power and Grounding:** VCC is connected to the SIM card and the PCB. GND is connected to the SIM card and the PCB. RST is connected to the SIM card and the PCB. VPP is connected to the SIM card and the PCB. CLK is connected to the SIM card and the PCB. I/O is connected to the SIM card and the PCB. GND1, GND2, GND3, and GND4 are connected to the SIM card and the PCB.
- Resistors:** R80 (15K) and R0402 are used for signal conditioning.
- Capacitors:** C68 (33pF), C69 (33pF), C70 (33pF), C0402, C71 (10uF), C72 (100nF), C0603, and C0402 are used for decoupling and timing.
- Diodes:** ED402, ED18, ED19, ED21, ED23, ED25, ED27, ED29, ED31, ED33, ED35, ED37, ED39, ED41, ED43, ED45, ED47, ED49, ED51, ED53, ED55, ED57, ED59, ED61, ED63, ED65, ED67, ED69, ED71, ED73, ED75, ED77, ED79, ED81, ED83, ED85, ED87, ED89, ED91, ED93, ED95, ED97, ED99 are used for signal protection and isolation.
- Header:** J4 is a 4-pin header connecting the SIM card to the PCB.

```

      <<PCIE30_PORT0_RXFLAP_IN
      <<PCIE30_PORT0_RXFLAP_IN
      <<PCIE30_PORT0_TX0P
      <<PCIE30_PORT0_TX0N
      <<PCIE30_PORT0_TX1P
      <<PCIE30_PORT0_TX1N
      <<PCIE30_PORT0_RX0P
      <<PCIE30_PORT0_RX0N
      <<PCIE30_PORT0_RX1P
      <<PCIE30_PORT0_RX1N
      <<PCIE30X4_WAKEN0_M1_L
      <<PCIE30X4_PERST0_M1_L
      <<PCIE30X4_CLKREQ0_M1_L
      >>PCIE_PWRDN_N
      >>PCIE_PWRDN_N
      >>SG_LTE_EN
      >>SG_LTE_EN

```



The schematic diagram illustrates the electrical connections for the M2 Key B H Socket. It shows the internal wiring of the socket, including the M2 Key B H Socket and the M2 Key B H Socket. The diagram includes various components such as resistors, capacitors, and connectors, with labels for pins and signals.

Key Components and Connections:

- M2 Key B H Socket:** The top section of the diagram shows the M2 Key B H Socket with pins 1 through 16. Key signals include:
 - CONF_2 = GND** (Pin 1)
 - CONF_1 = NC** (Pin 2)
 - CONF_1 = NC** (Pin 3)
 - CONF_1 = NC** (Pin 4)
 - CONF_1 = NC** (Pin 5)
 - CONF_1 = NC** (Pin 6)
 - CONF_1 = NC** (Pin 7)
 - CONF_1 = NC** (Pin 8)
 - CONF_1 = NC** (Pin 9)
 - CONF_1 = NC** (Pin 10)
 - CONF_1 = NC** (Pin 11)
 - CONF_1 = NC** (Pin 12)
 - CONF_1 = NC** (Pin 13)
 - CONF_1 = NC** (Pin 14)
 - CONF_1 = NC** (Pin 15)
 - CONF_1 = NC** (Pin 16)
- M2 Key B H Socket:** The bottom section of the diagram shows the M2 Key B H Socket with pins 1 through 16. Key signals include:
 - CONF_1 = NC** (Pin 1)
 - CONF_1 = NC** (Pin 2)
 - CONF_1 = NC** (Pin 3)
 - CONF_1 = NC** (Pin 4)
 - CONF_1 = NC** (Pin 5)
 - CONF_1 = NC** (Pin 6)
 - CONF_1 = NC** (Pin 7)
 - CONF_1 = NC** (Pin 8)
 - CONF_1 = NC** (Pin 9)
 - CONF_1 = NC** (Pin 10)
 - CONF_1 = NC** (Pin 11)
 - CONF_1 = NC** (Pin 12)
 - CONF_1 = NC** (Pin 13)
 - CONF_1 = NC** (Pin 14)
 - CONF_1 = NC** (Pin 15)
 - CONF_1 = NC** (Pin 16)

The diagram also shows various other components and connections, including resistors, capacitors, and connectors, with labels for pins and signals.

Option:

No used, Just reserve

PCIE30X2_CLKREQM_M3_JV3 R620 1 2 22n 5k PCIE30X2_CLKREQM_M3_JV3 R6402

PCIE30X2_WAKEN_M3_JV3 R621 1 2 22n 5k PCIE30X2_WAKEN_M3_JV3 R6402

PCIE30X2_PERRSTN_M3_JV3 R622 1 2 22n 5k PCIE30X2_PERRSTN_M3_JV3 R6402

PCIE30 REFCLKP COM1 PCIE30 REFCLKP COM1

PCIE30 PORT1_TXP C512 1 2 220nF C0201 X58 10V PCIE30 TX2 P PCIE30 PORT1_TX2N C513 1 2 220nF C0201 X58 10V PCIE30 TX2 N

PCIE30 PORT1_TX3P C514 1 2 220nF C0201 X58 10V PCIE30 TX3 P PCIE30 PORT1_TX3N C515 1 2 220nF C0201 X58 10V PCIE30 TX3 N

PCIE30 PORT1_RX2P PCIE30 PORT1_RX2N

VCCJV3_PCIE30 VCCJV3_PCIE30

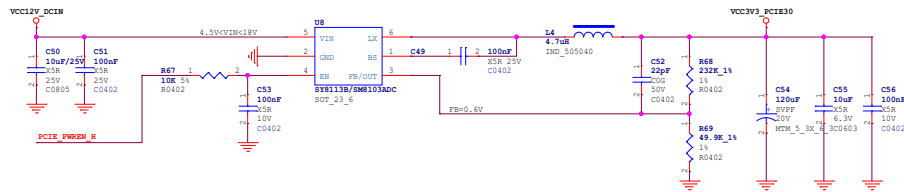
C516 22uF C517 100nF C518 22uF C519 100nF

X58 X58 X58 X58

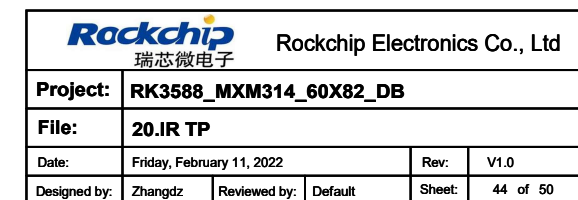
6.3V 10V 6.3V 10V

C0603 C4002 C0603 C4002

M.2 Power

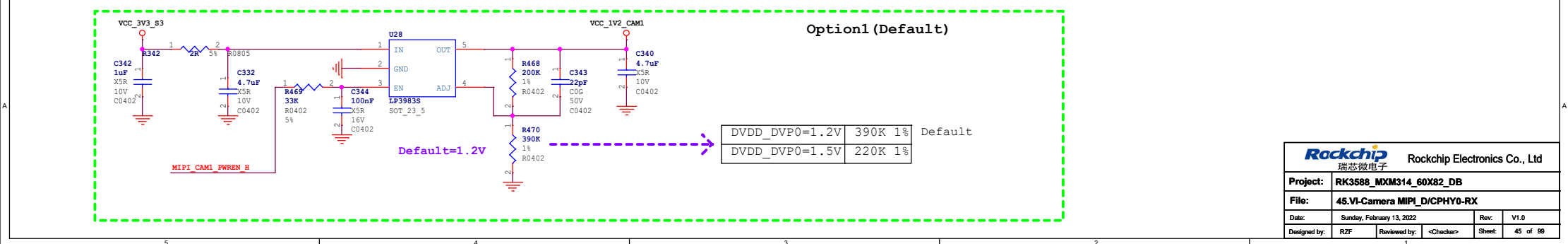
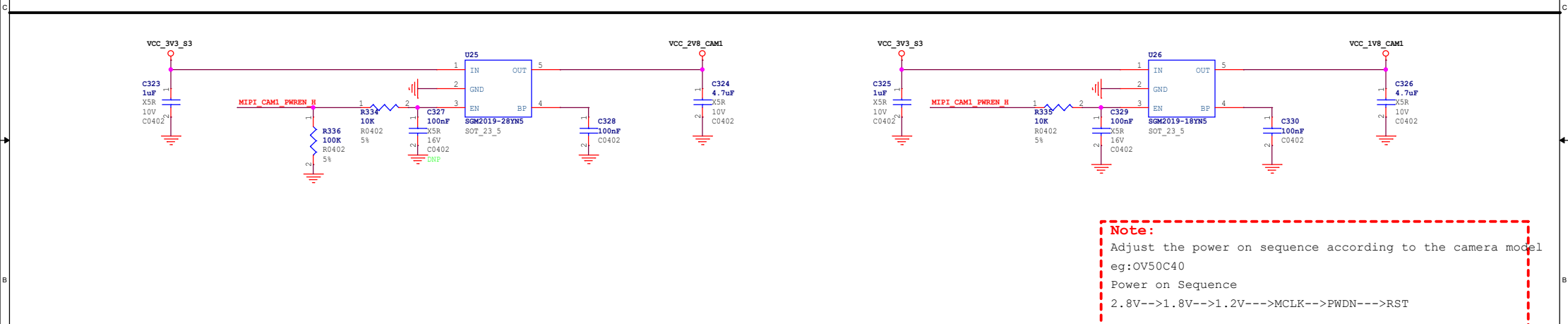
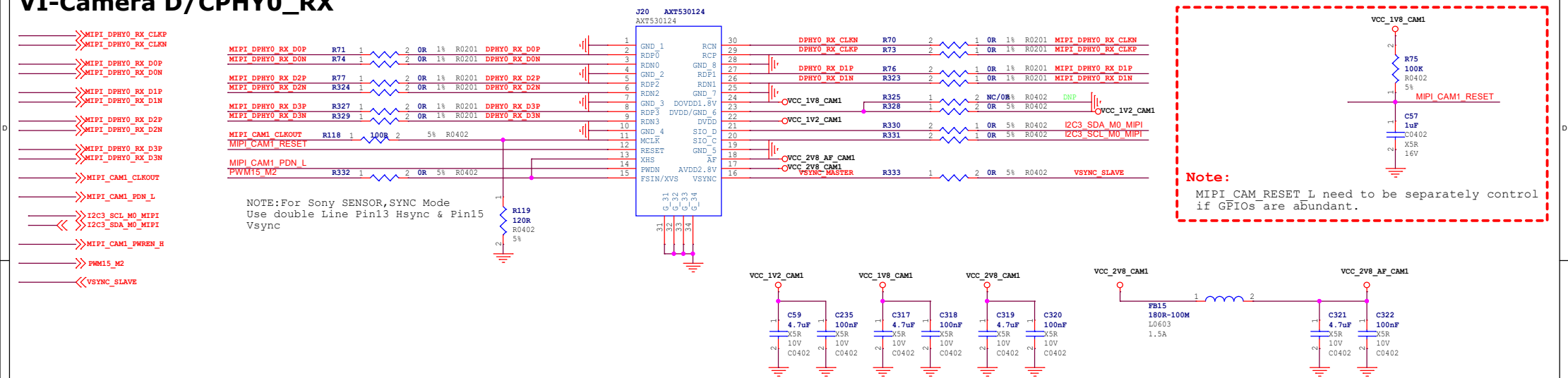


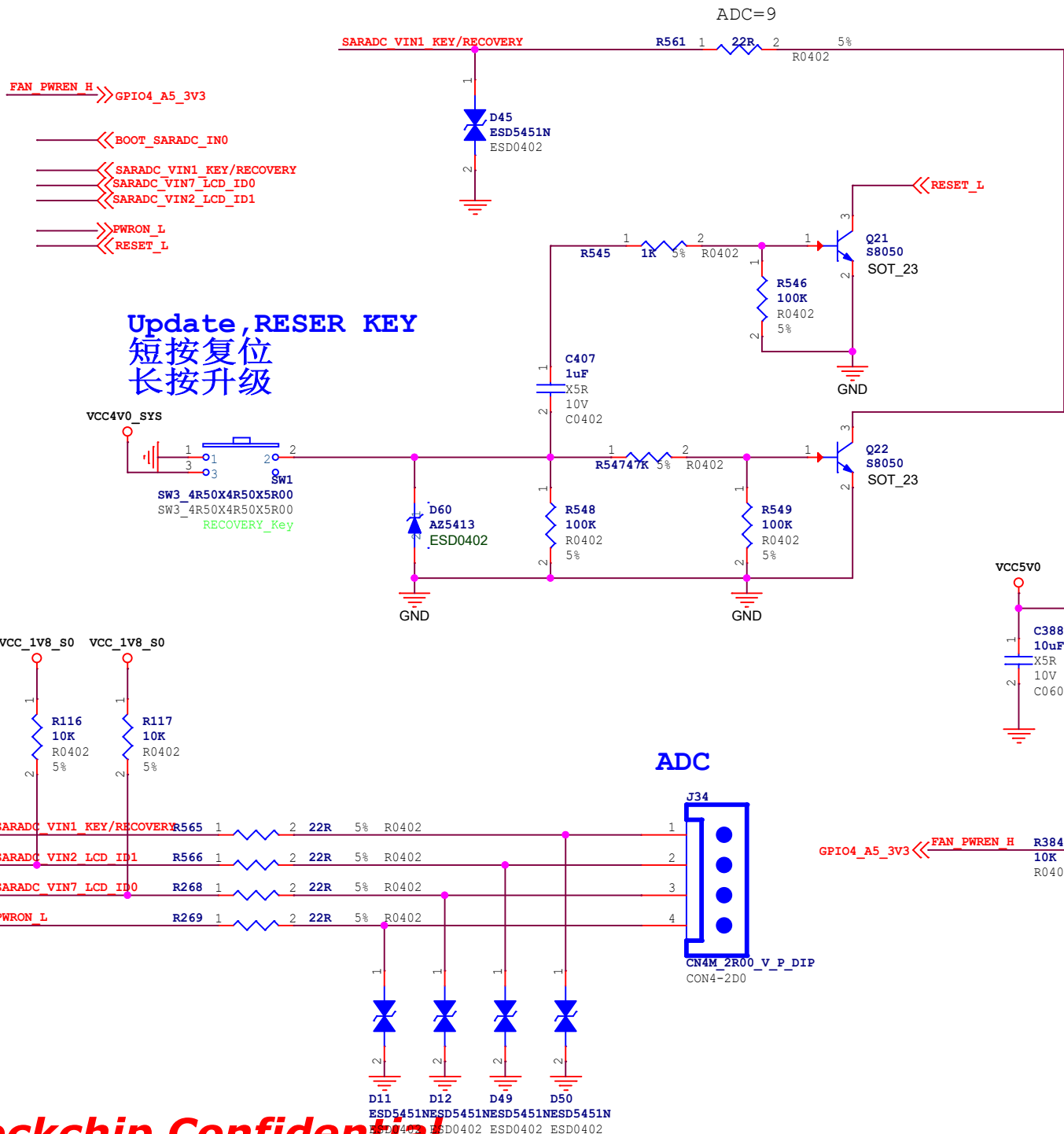
Default 3.39V



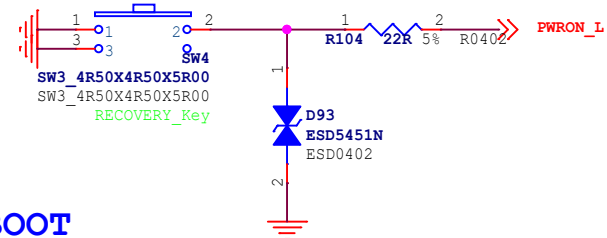
5	4	3	2	1
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VI-Camera D/CPHY0_RX

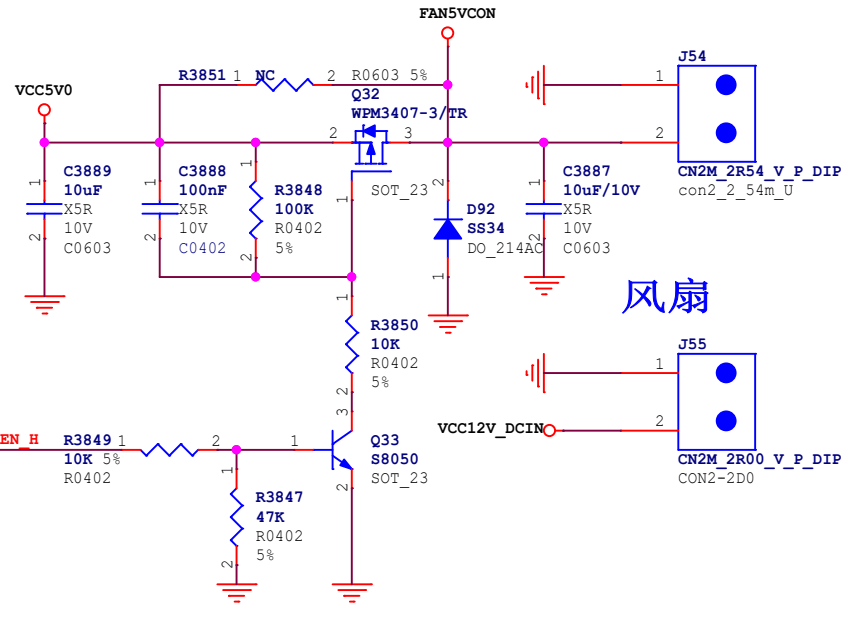
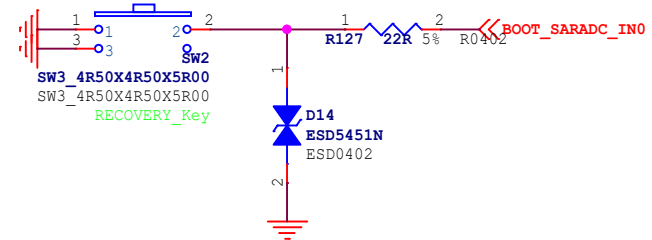




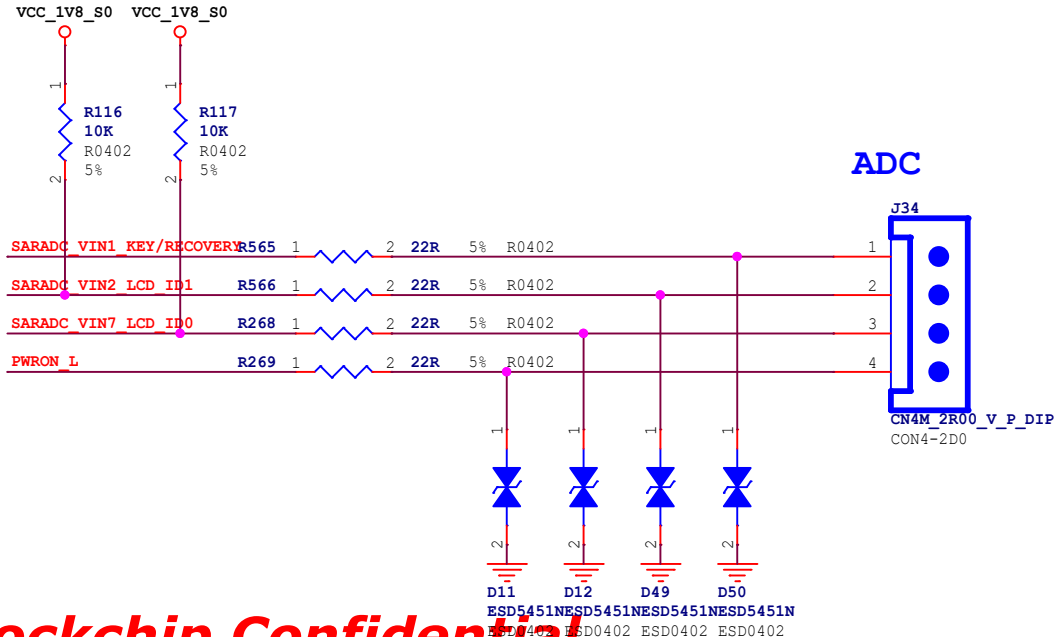
POWER KEY



BOOT

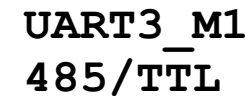


ADC



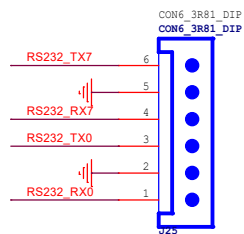
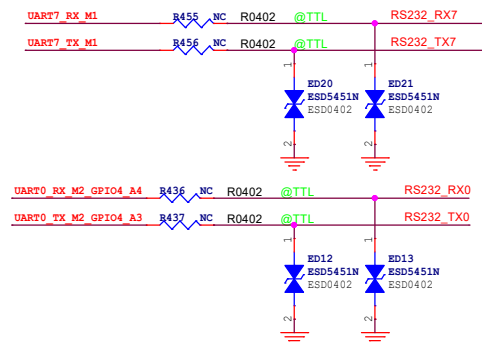
Rockchip Confidential

 瑞芯微电子		Rockchip Electronics Co., Ltd	
Project:	RK3588_MXM314_60X82_DB		
File:	21.KEY Array		
Date:	Friday, January 28, 2022		Rev: V1.0
Designed by:	Zhangdz	Reviewed by: Default	Sheet: 46 of 50



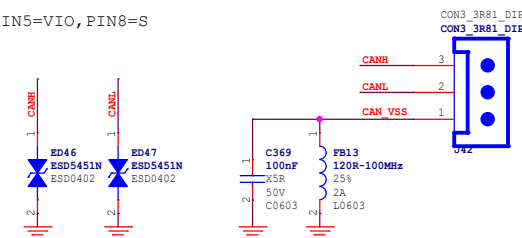
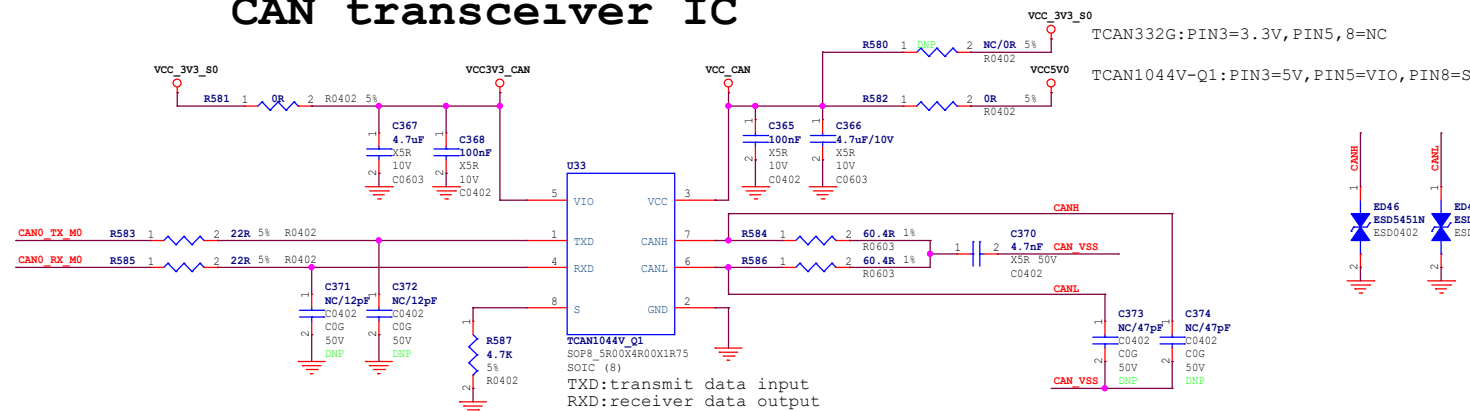
Schematic diagram of the RS485 module circuit. The circuit includes a MAX3485EESD (U19) transceiver, a 3.3V regulator (U18), and various passive components like resistors (R158, R159, R162, R0402) and capacitors (C130, C0402). It shows connections for VCC, GND, TX, RX, and RS485_A/B signals. A note indicates that the module is not tested for RS485+/-12V.

UART0 M2

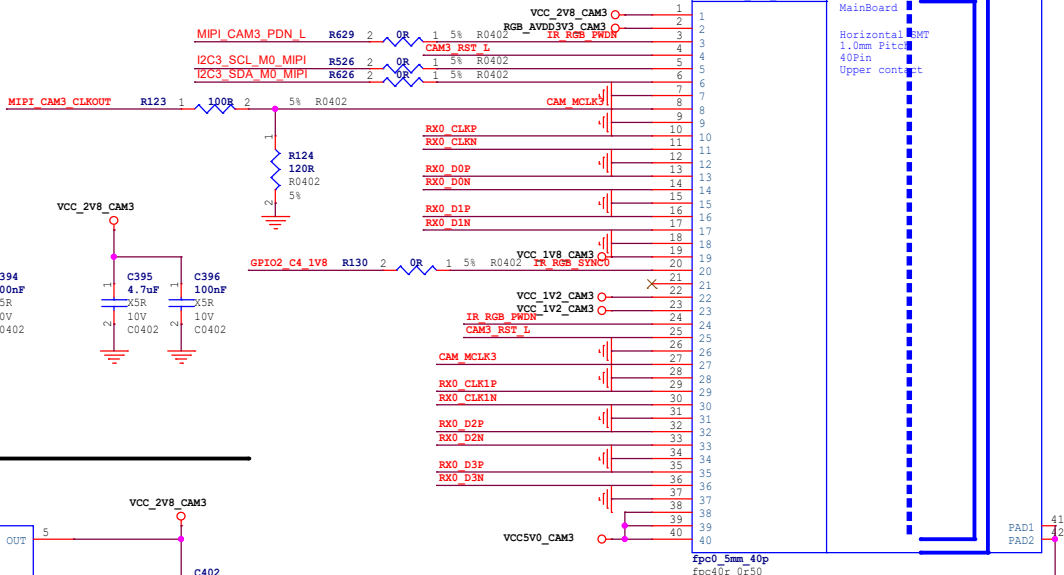
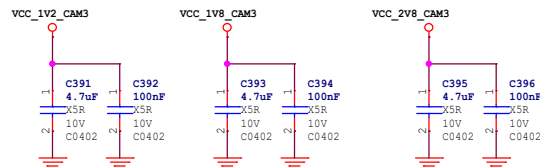
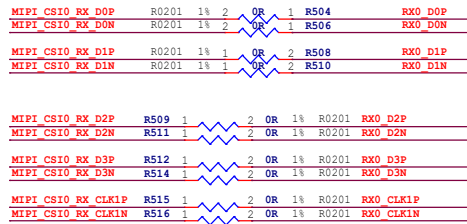


CAN0 M0

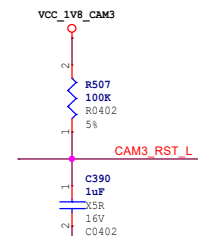
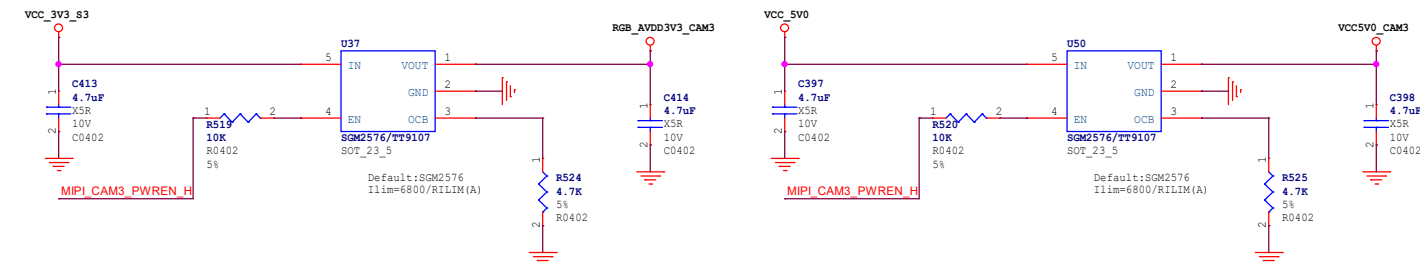
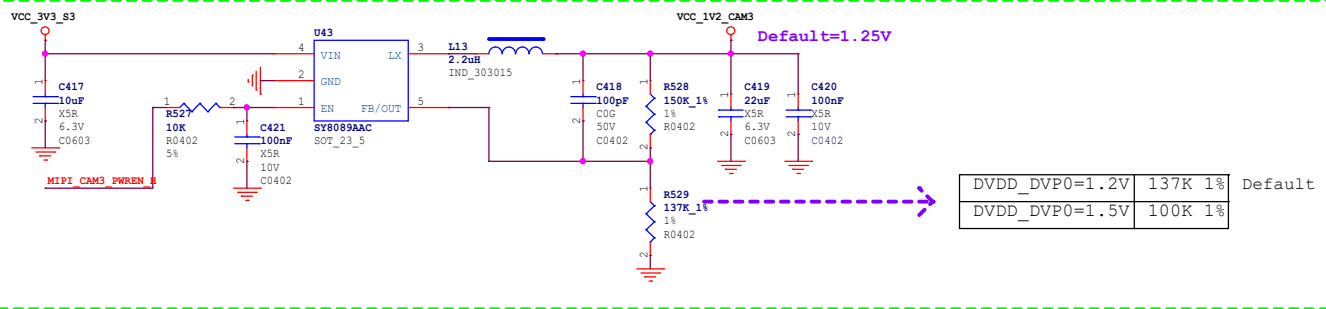
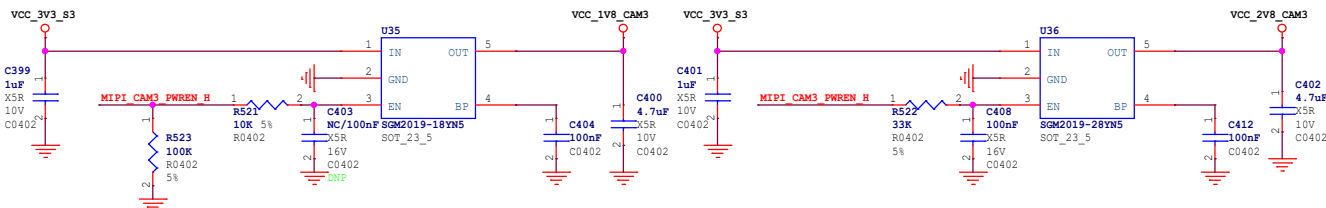
CAN transceiver IC



MIP1 CS10 RX CLK0P	R0201	1	1	0R	2	R505	RX0 CLKP
MIP1 CS10 RX CLK0N	R0201	1	1	0R	2	R503	RX0 CLKN
MIP1 CS10 RX D0P	R0201	1	2	0R	1	R504	RX0 D0P
MIP1 CS10 RX D0N	R0201	1	2	0R	1	R506	RX0 D0N
MIP1 CS10 RX D1P	R0201	1	1	0R	2	R508	RX0 D1P
MIP1 CS10 RX D1N	R0201	1	1	0R	2	R510	RX0 D1N

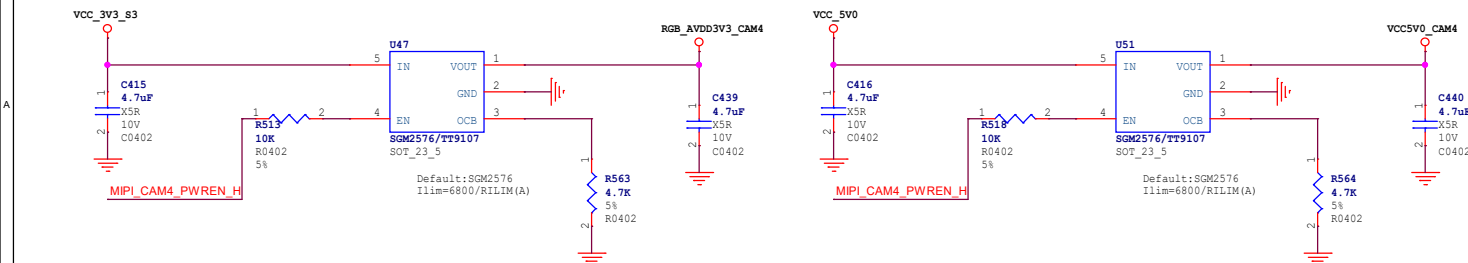
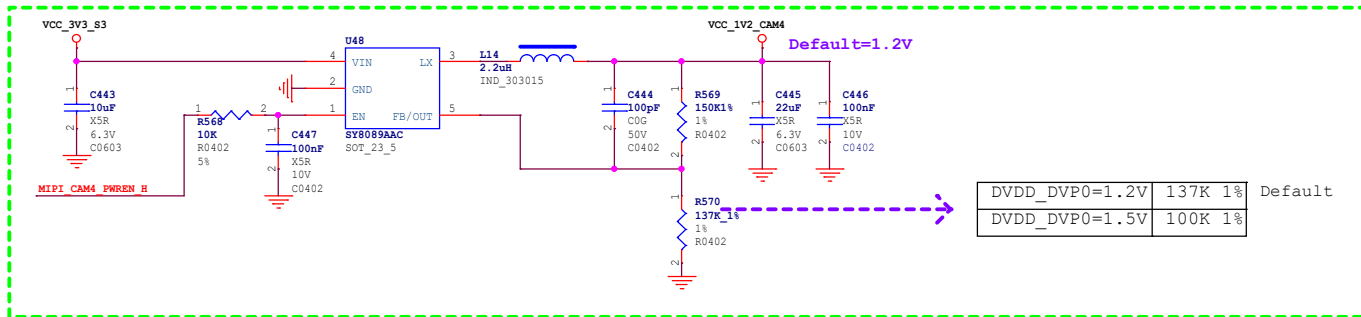
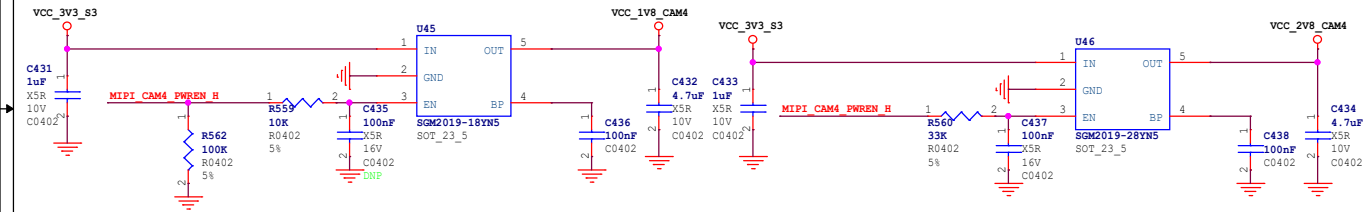
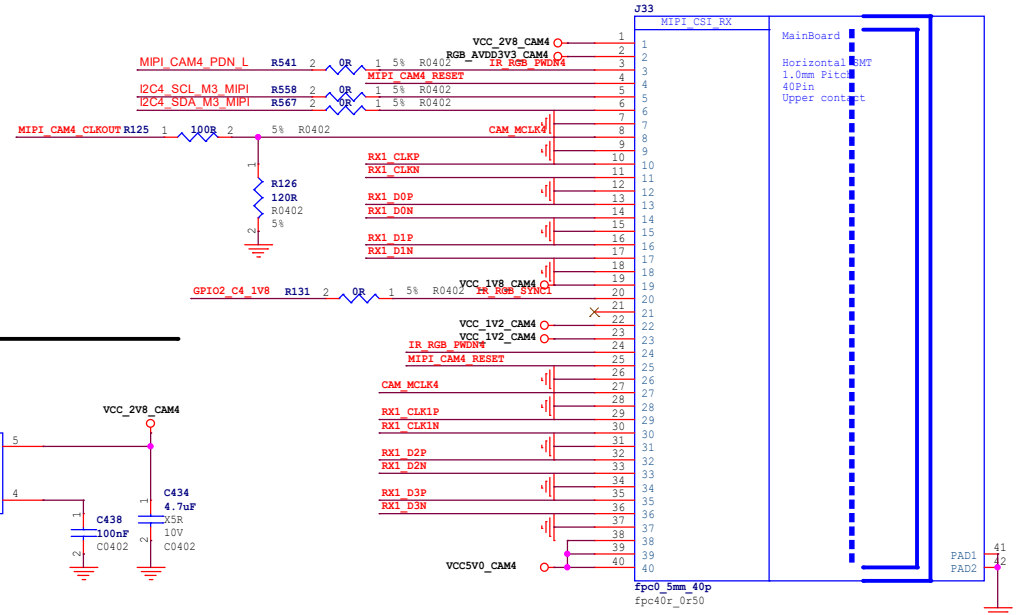
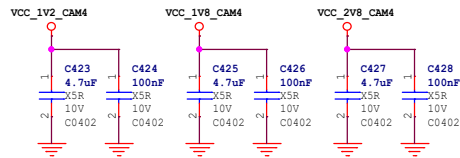
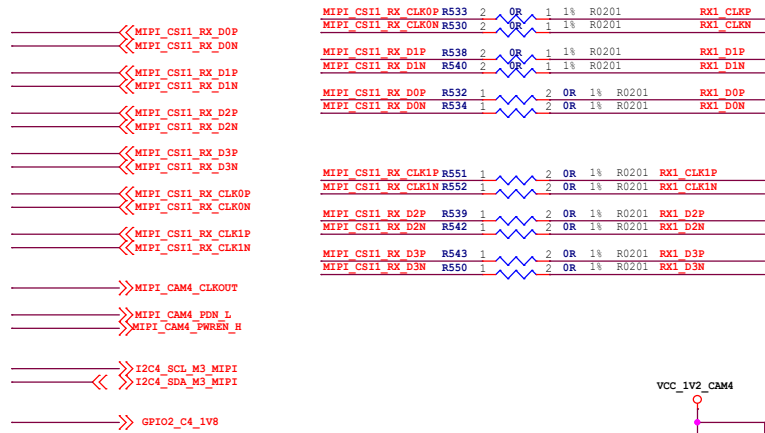


1.8V-->1.2V-->2.8V--->MCLK-->PWDN--->RST



MIPI CAM RESET L need to be separately control
if GPIOs are abundant.

MIPI-DPHY1_RX

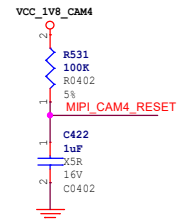


Note:

Adjust the power on sequence according to the camera model
eg:GC8034

Power on Sequence

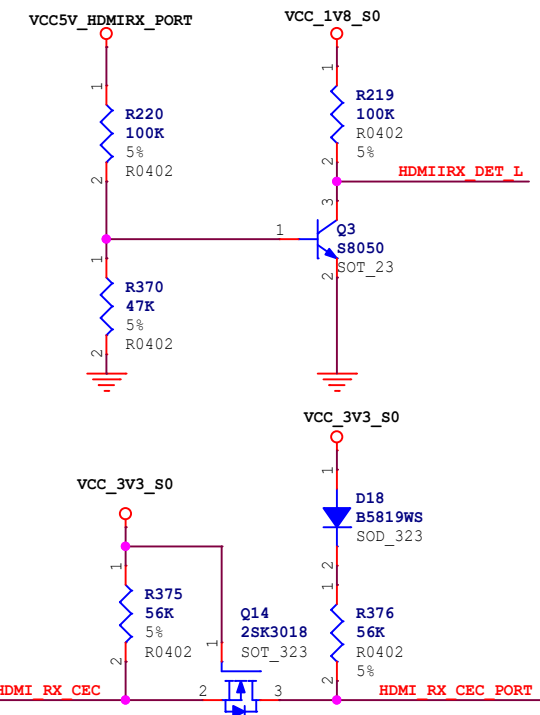
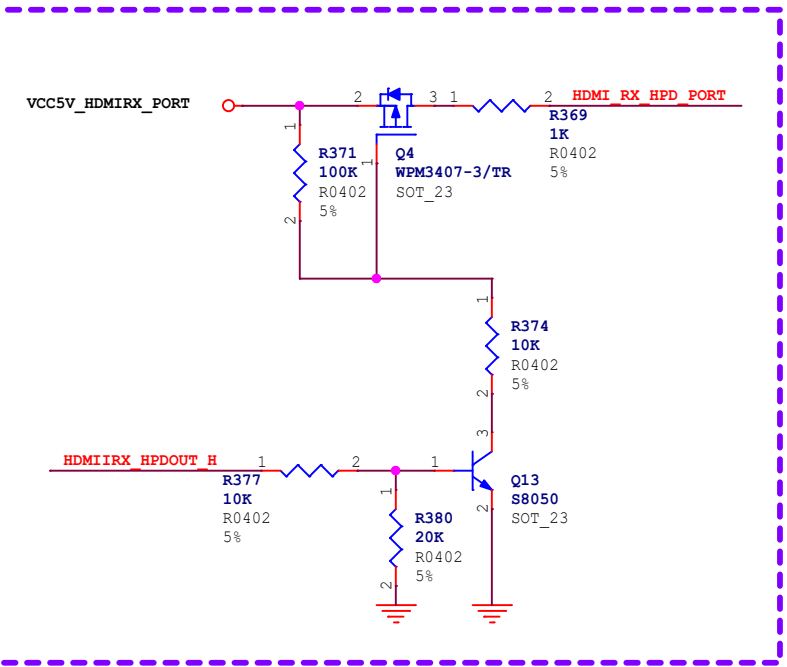
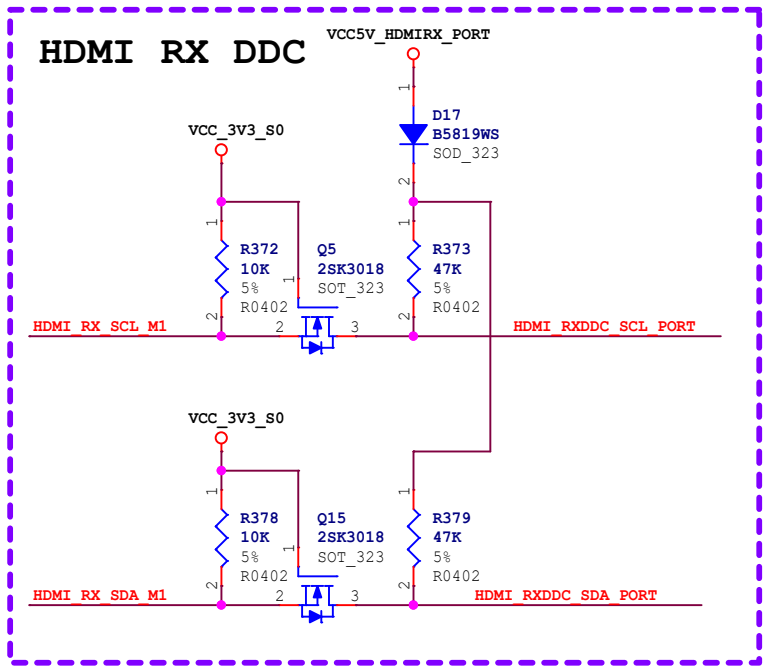
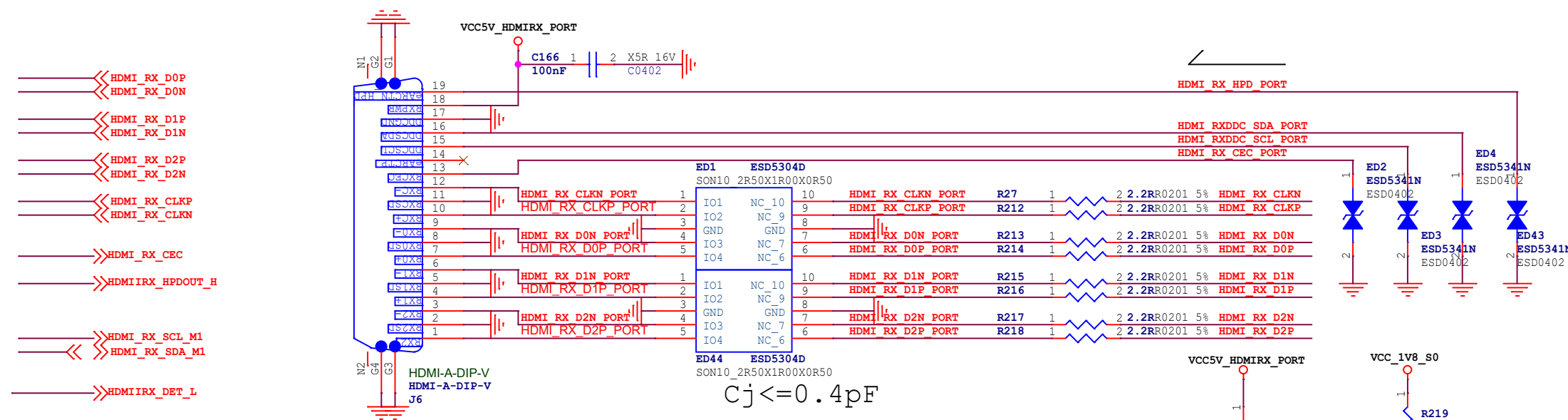
1.8V-->1.2V-->2.8V--->MCLK-->PWDN--->RST



Note:

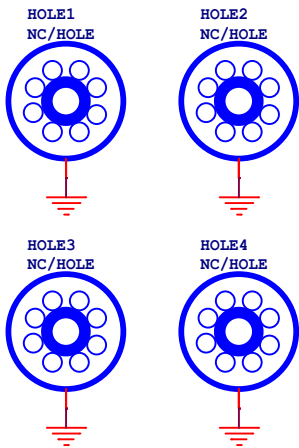
MIPI_CAM_RESET_L need to be separately control if GPIOs are abundant.

HDMI2.0 RX

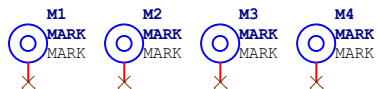


 瑞芯微电子		Rockchip Electronics Co., Ltd	
Project:	RK3588_MXM314_60X82_DB		
File:	06.VI-HDMI2.0 RX		
Date:	Friday, February 11, 2022		Rev: V1.0
Designed by:	RZF	Reviewed by:	Sheet: 49 of 99

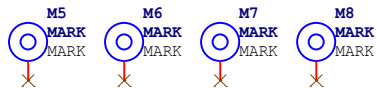
HPD:
Sink Side: Require output, Min 2.4V; Max 5.3V
Source Side: Require input and Detection.
Min 2.0V, Max: 5.3V



TOP Mark

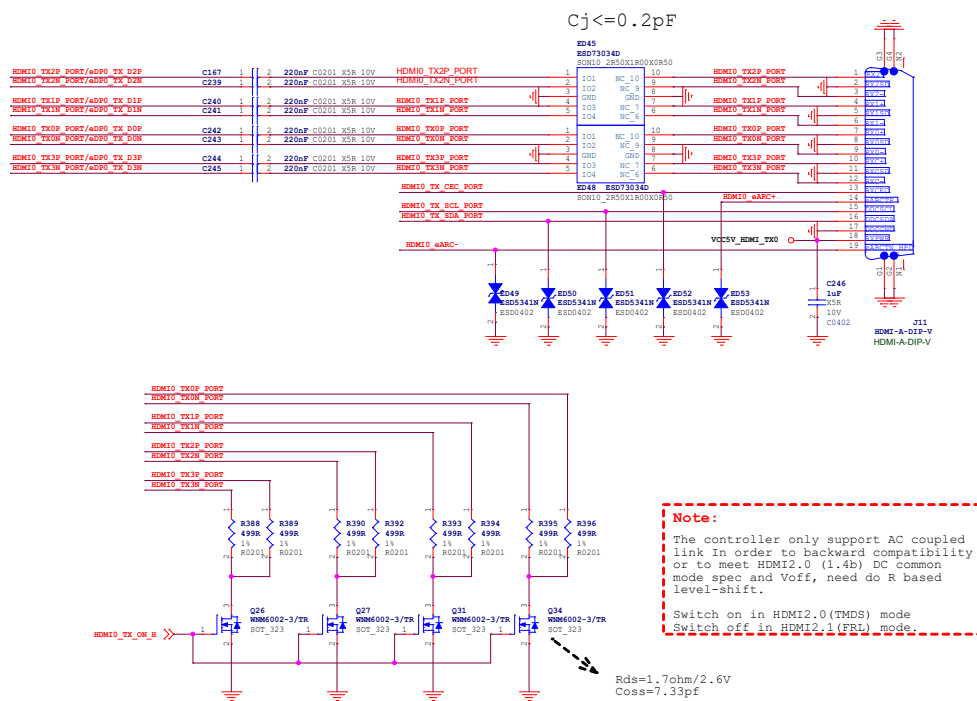
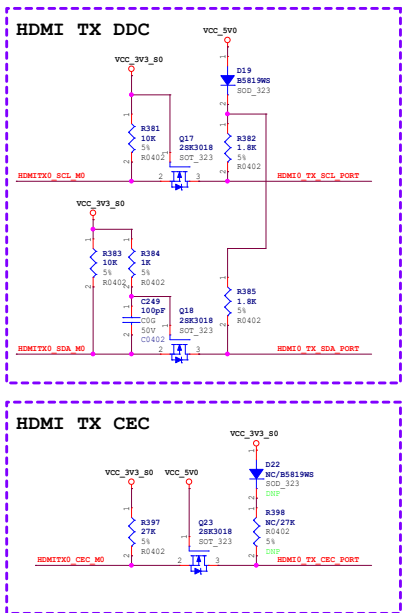
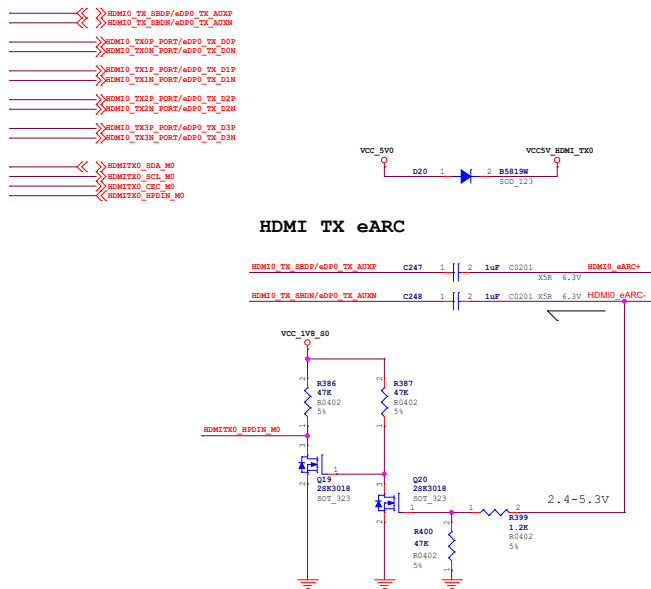


BOTTOM Mark

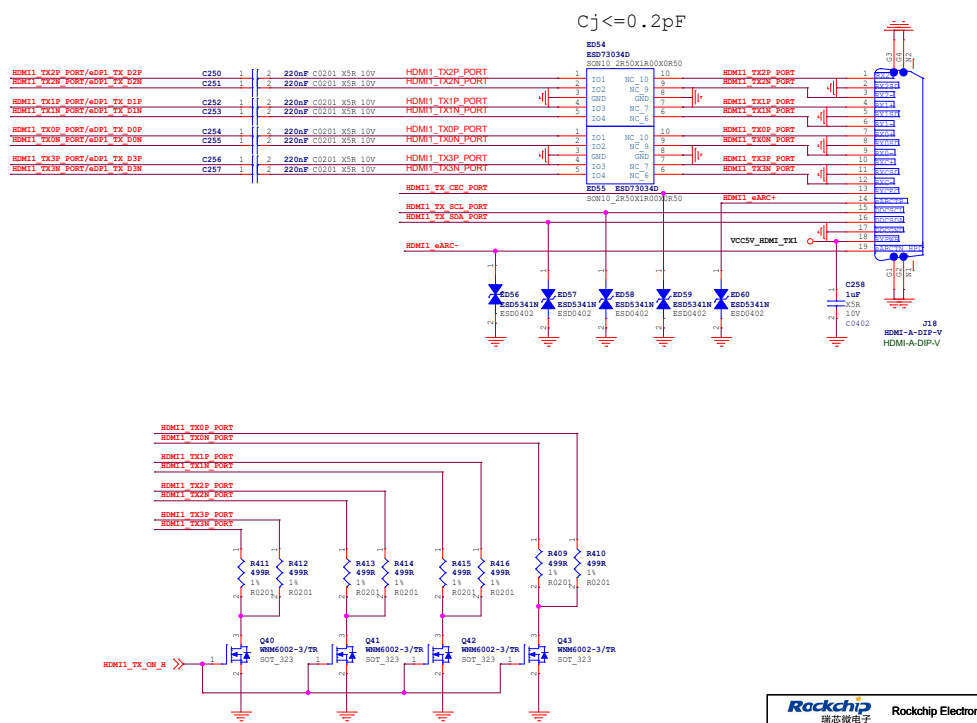
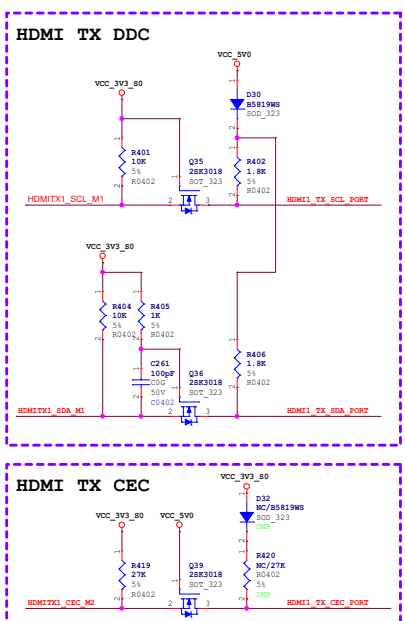
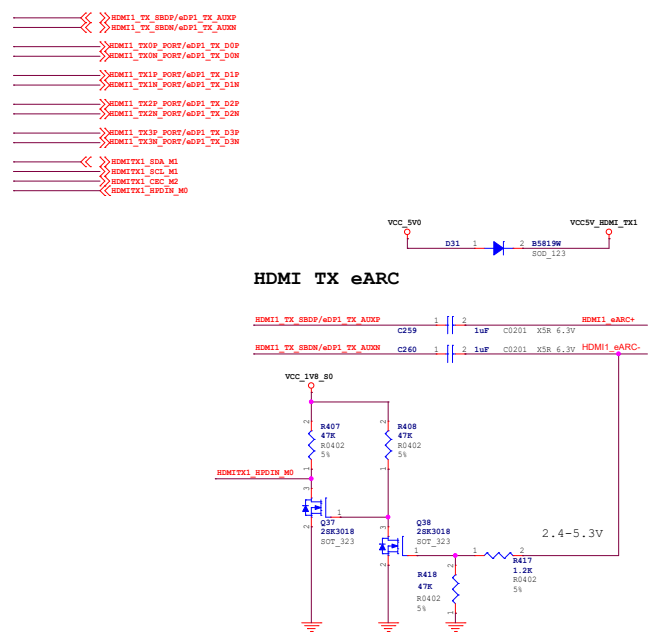


Rockchip 瑞芯微电子 Rockchip Electronics Co., Ltd				
Project:	RK3588_MXM314_60X82_DB			
File:	99.Mark/Hole/Heatsink			
Date:	Friday, January 28, 2022		Rev:	V1.0
Designed by:	Zhangdz	Reviewed by:	Default	Sheet: 50 of 50

HDMI TX0

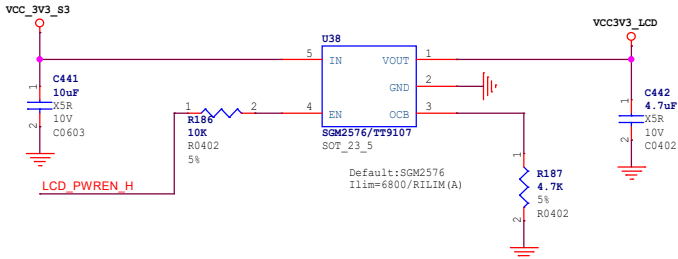


HDMI TX1

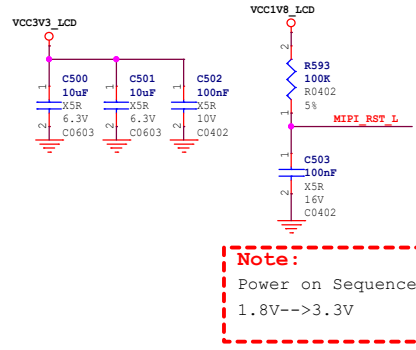
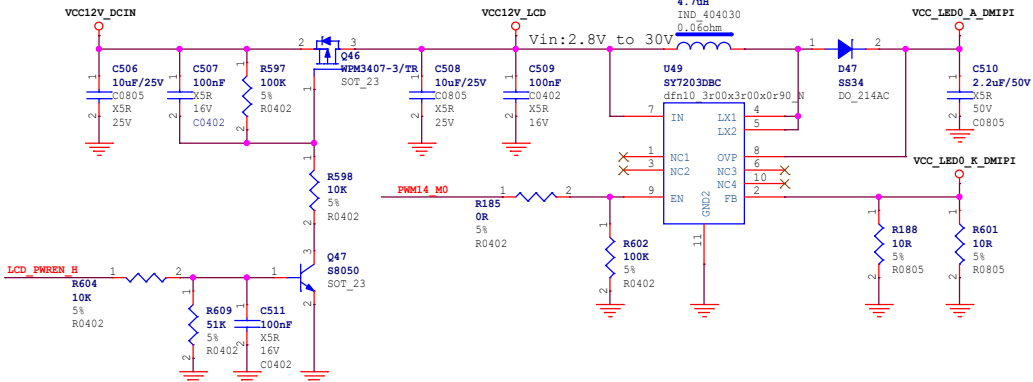
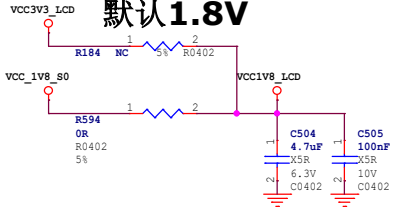


Dual-MIPI LCM

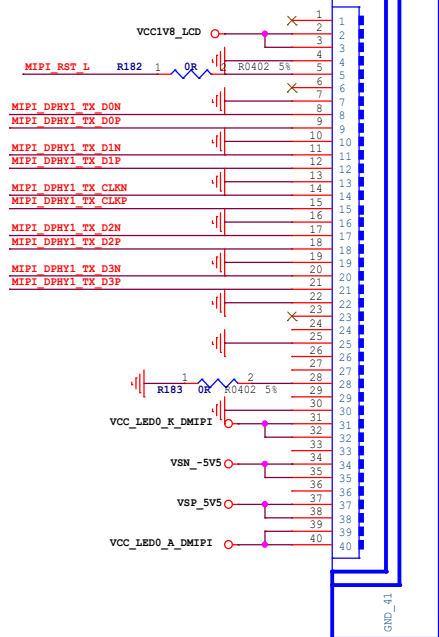
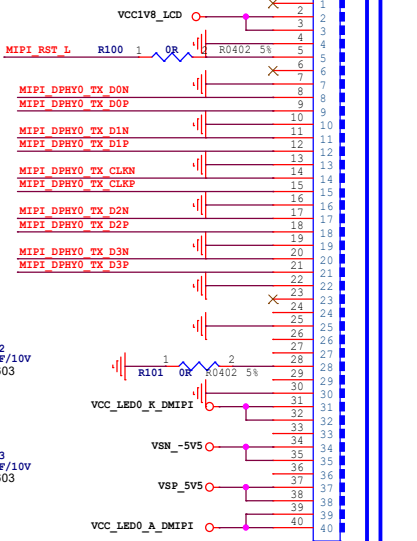
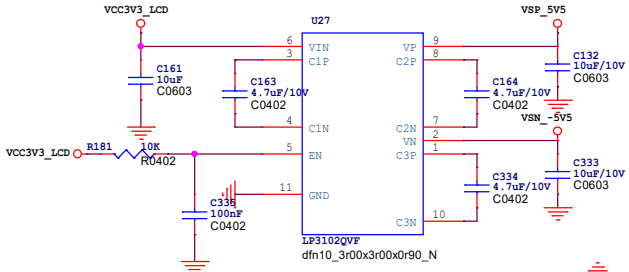
- >>>MIPI_DPHY0_TX_CLKP
- >>>MIPI_DPHY0_TX_CLKN
- >>>MIPI_DPHY0_TX_D0P
- >>>MIPI_DPHY0_TX_D0N
- >>>MIPI_DPHY0_TX_D1P
- >>>MIPI_DPHY0_TX_D1N
- >>>MIPI_DPHY0_TX_D2P
- >>>MIPI_DPHY0_TX_D2N
- >>>MIPI_DPHY0_TX_D3P
- >>>MIPI_DPHY0_TX_D3N
- >>>MIPI_DPHY1_TX_CLKP
- >>>MIPI_DPHY1_TX_CLKN
- >>>MIPI_DPHY1_TX_D0P
- >>>MIPI_DPHY1_TX_D0N
- >>>MIPI_DPHY1_TX_D1P
- >>>MIPI_DPHY1_TX_D1N
- >>>MIPI_DPHY1_TX_D2P
- >>>MIPI_DPHY1_TX_D2N
- >>>MIPI_DPHY1_TX_D3P
- >>>MIPI_DPHY1_TX_D3N
- >>>PWM14_M0
- >>>GPIO3_C2_3V3
- >>>LCD_PWREN_H



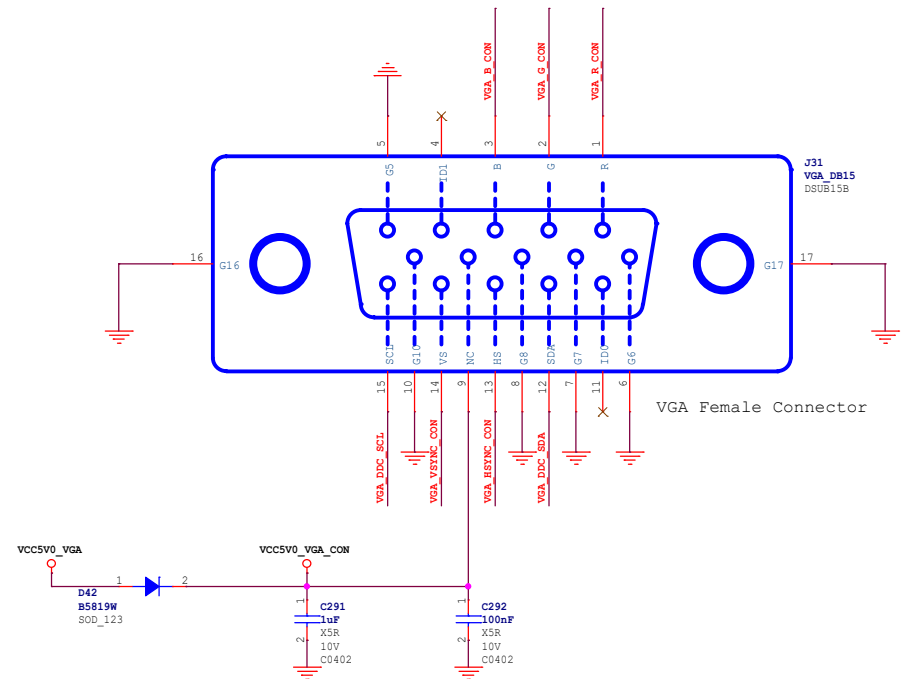
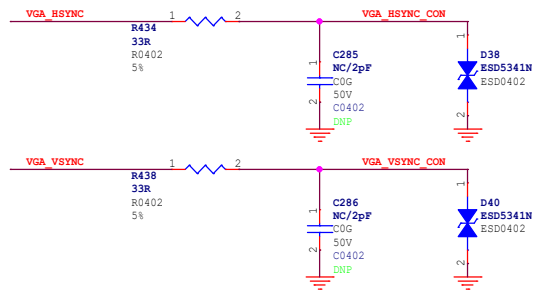
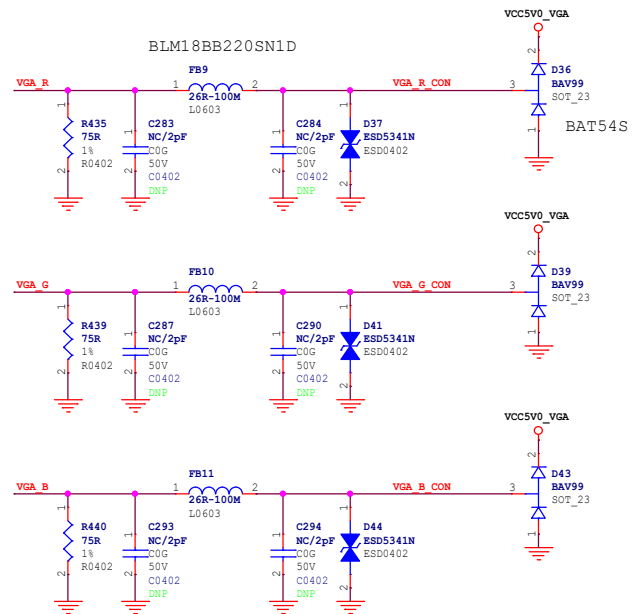
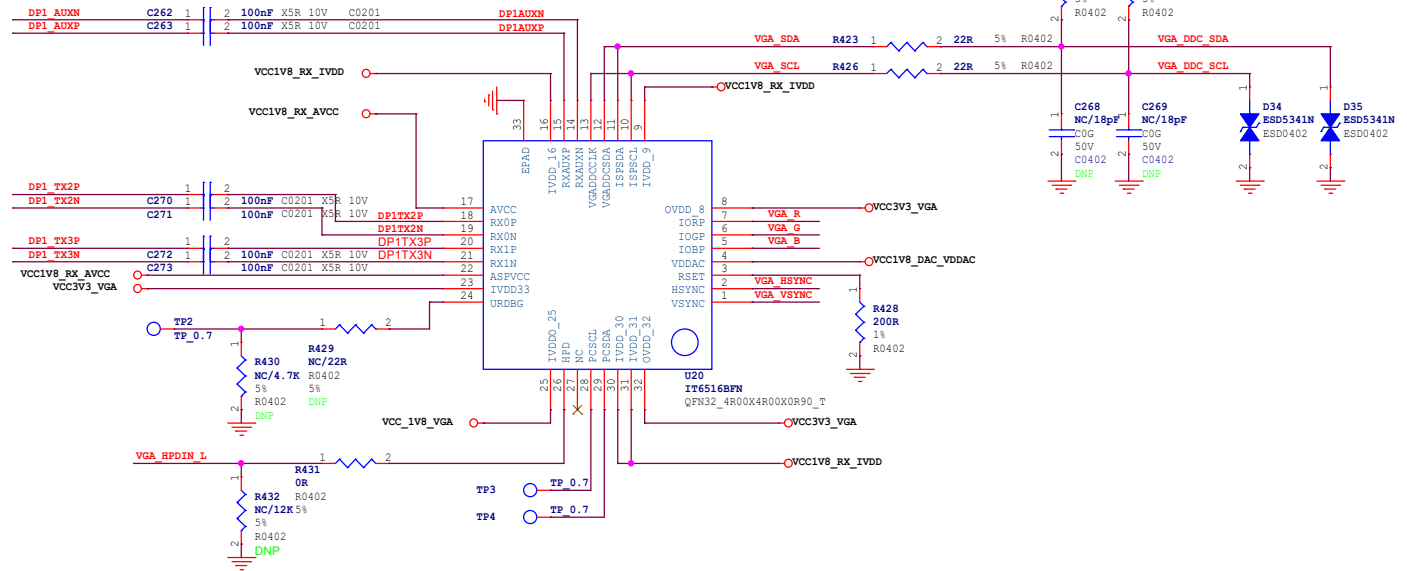
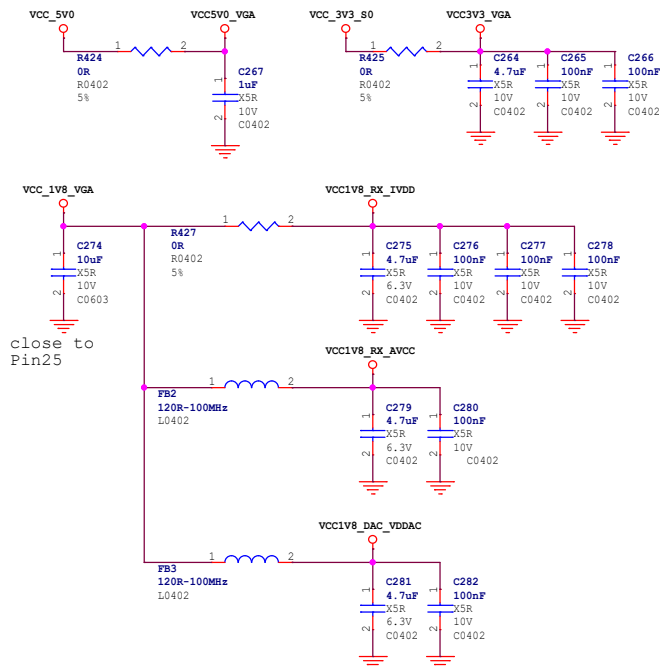
默认1.8V



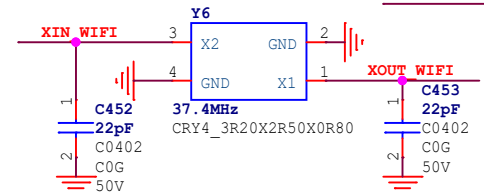
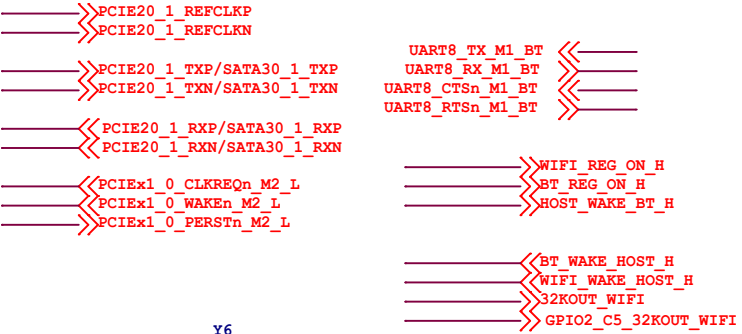
Note:
Power on Sequence
1.8V-->3.3V



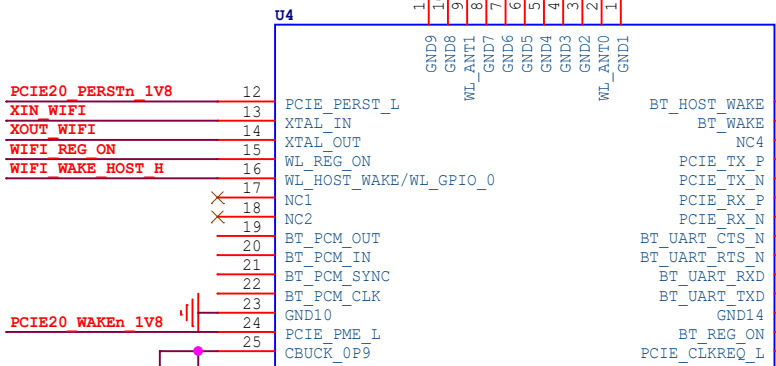
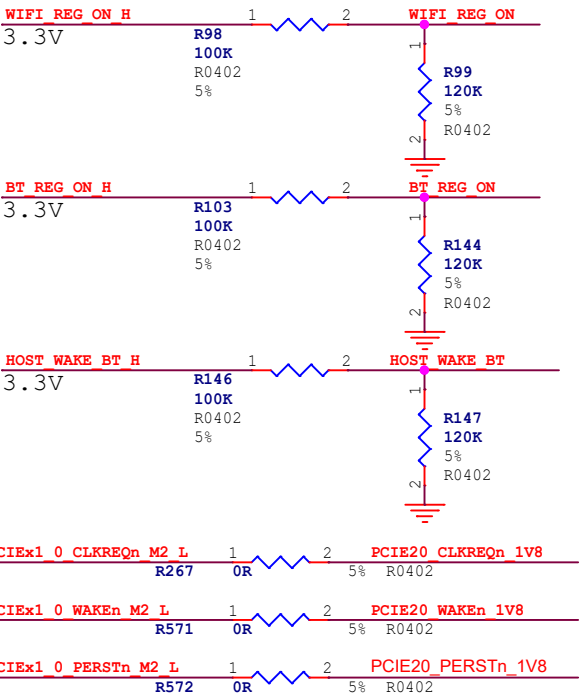
VGA OUTPUT



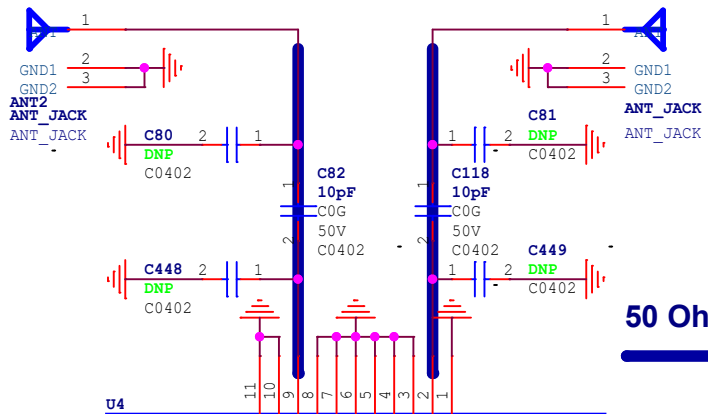
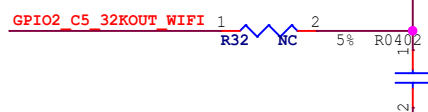
PCIE WIFI6/BT Module-2T2R



NOTE:
Adjust the load capacitor
according to the crystal spec.

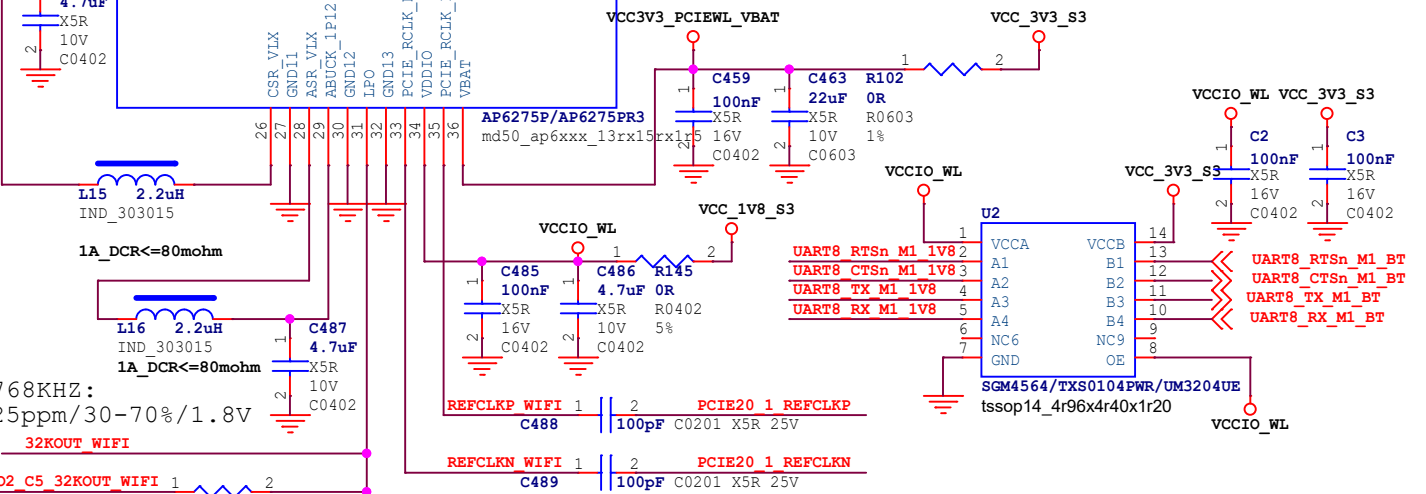
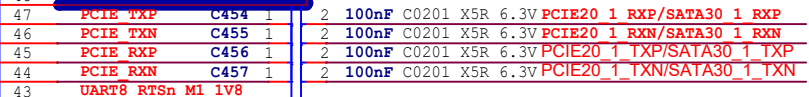


32.768KHZ:
+/-25ppm/30-70%/1.8V
32KOUT_WIFI



50 Ohm RF trace

This standalone BT-ANT is reserved for AP6275PR3. Leave PIN48 float for AP6275P, of which BT-ANT is mux with WIFI.



VBAT: (3.1-3.8V) / 1.2A
VDDIO: (1.68-1.98V) / 300mA.
WIFI: a/b/g/n/ac/ax 2T2R MIMO
BT: 5.0

 瑞芯微电子		Rockchip Electronics Co., Ltd				
Project:	RK3588_MXM314_60X82_DB					
File:	63.WIFI/BT-PCIe_2T2R					
Date:	Wednesday, March 23, 2022				Rev:	V1.0
Designed by:	RZF	Reviewed by:		Sheet:	63 of 99	